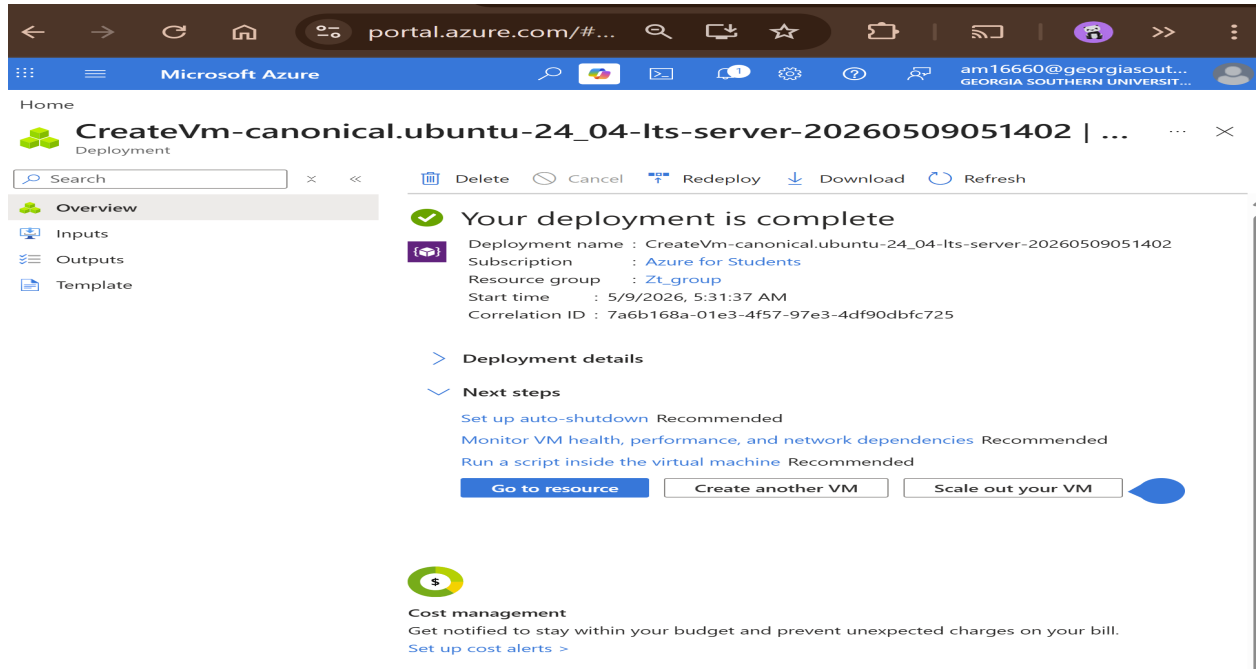


Appendix A. Screenshot Evidence Packet

1. This screenshot shows that the Azure Ubuntu virtual machine deployment completed successfully.

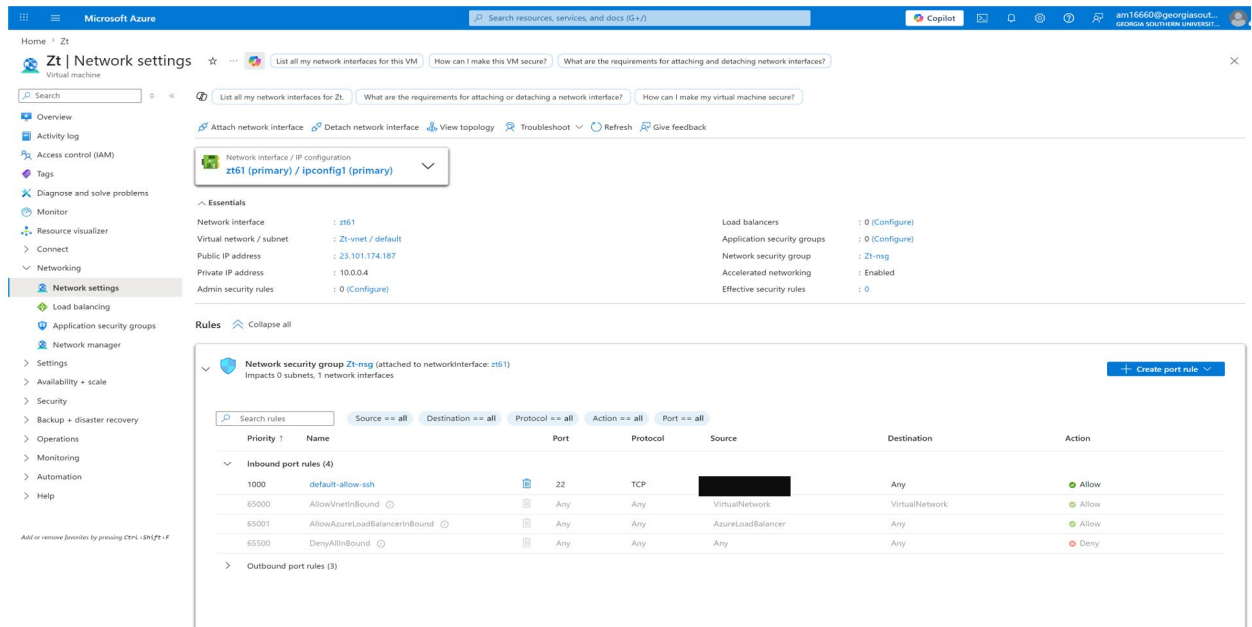


2. This screenshot shows the cloud resources created for the lab, including the virtual machine, network interface, network security group, virtual network, and public IP address.

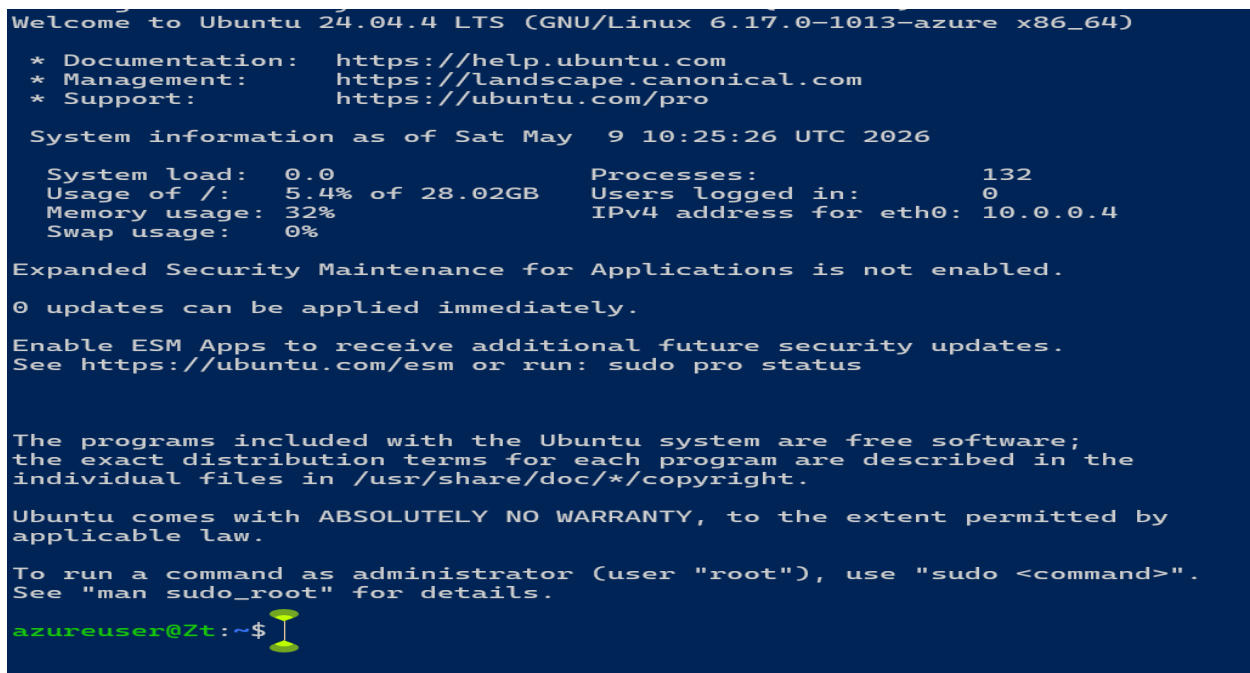
Deployment details

	Resource	Type	Status
✓	Zt	Microsoft.Compute/virtualMach	OK
✓	zt61	Microsoft.Network/networkInter	OK
✓	Zt-nsg	Network security group	OK
✓	Zt-vnet	Virtual network	OK
✓	Zt-ip	Public IP address	OK

- The SSH inbound rule was restricted from "Any" source to a specific trusted public IP address using /32 notation. This reduced unnecessary exposure of port 22 while still allowing secure remote administration.



- This screenshot confirms successful SSH access to the Azure Ubuntu VM.



5. The system update check confirmed that the running kernel was up to date and that no services required restart.

```
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@Zt:~$ nmap --version
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.6 openssl-3.0.13 libssh2-1.11.0 libz-1.3 libpcap-1.10.4
nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
azureuser@Zt:~$ tcpdump --version
tcpdump version 4.99.4
libpcap version 1.10.4 (with TPACKET_V3)
OpenSSL 3.0.13 20 Jan 2024
azureuser@Zt:~$
```

6. This screenshot shows the creation of the project evidence folder, a simple protected test resource, and a Python HTTP server running on port 8080. The `curl` command confirms that the VM successfully served the protected resource from localhost.

```
azureuser@Zt:~$ mkdir -p ~/zt_lab/evidence ~/zt_lab/www
azureuser@Zt:~$ echo "Zero Trust protected test resource - Ubuntu VM" > ~/zt_lab/www/index.html
azureuser@Zt:~$ cd ~/zt_lab/www
python3 -m http.server 8080 > ~/zt_lab/evidence/http_server.log 2>&1 &
[1] 3595
azureuser@Zt:~/zt_lab/www$ curl http://127.0.0.1:8080
Zero Trust protected test resource - Ubuntu VM
azureuser@Zt:~/zt_lab/www$
```

7. This screenshot shows **tcpdump** capturing traffic on the Azure Ubuntu VM while controlled HTTP traffic was generated against the protected resource on port 8080. The evidence directory confirms that the HTTP server log and packet capture file were created successfully.

```
azureuser@Zt:~/zt_lab/www$ sudo timeout 30 tcpdump -i any -nn 'tcp port 8080 or icmp' -w ~/zt_lab/evidence/zt_capture.pcap &
[2] 3653
azureuser@Zt:~/zt_lab/www$ libibverbs: Warning: couldn't open config directory '/etc/libibverbs.d'.
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
curl http://127.0.0.1:8080
Zero Trust protected test resource - Ubuntu VM
azureuser@Zt:~/zt_lab/www$ nmap -sV -p 8080 127.0.0.1 | tee ~/zt_lab/evidence/nmap_8080.txt12 packets captured
26 packets received by filter
0 packets dropped by kernel
ls -lh ~/zt_lab/evidence
total 8.0K
-rw-rw-r-- 1 azureuser azureuser 240 May  9 10:48 http_server.log
-rw-r--r-- 1 tcpdump  tcpdump  1.4K May  9 10:48 zt_capture.pcap
[2]+  Exit 124                  sudo timeout 30 tcpdump -i any -nn 'tcp port 8080 or icmp' -w ~/zt_lab/evidence/zt_capture.pcap
```

8. This screenshot shows readable packet-capture output from the saved **.pcap** file. The output includes TCP traffic to port 8080, an HTTP GET request, and a successful HTTP/1.0 200 OK response, confirming that the VM generated and monitored real network traffic.

```
azureuser@Zt:~/zt_lab/www$ sudo tcpdump -nn -r ~/zt_lab/evidence/zt_capture.pcap | head -20
reading from file /home/azureuser/zt_lab/evidence/zt_capture.pcap, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144
Warning: interface names might be incorrect
10:48:14.168227 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [S], seq 1320977220, win 65495, options [mss 65495,sackOK,TS val 2197598675 ecr 0,nop,wscale 7], length 0
10:48:14.168237 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [S.], seq 2833041329, ack 1320977221, win 65483, options [mss 65495,sackOK,TS val 2197598675 ecr 2197598675,nop,wscale 7], length 0
10:48:14.168248 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [.], ack 1, win 512, options [nop,nop,TS val 2197598675 ecr 2197598675], length 0
10:48:14.168307 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [P.], seq 1:78, ack 1, win 512, options [nop,nop,TS val 2197598675 ecr 2197598675], length 77: HTTP: GET / HTTP/1.1
10:48:14.168314 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [.], ack 78, win 511, options [nop,nop,TS val 2197598675 ecr 2197598675], length 0
10:48:14.168862 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [P.], seq 1:186, ack 78, win 512, options [nop,nop,TS val 2197598676 ecr 2197598675], length 185: HTTP: HTTP/1.0 200 OK
10:48:14.168880 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [.], ack 186, win 511, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0
10:48:14.168912 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [P.], seq 186:233, ack 78, win 512, options [nop,nop,TS val 2197598676 ecr 2197598676], length 47: HTTP
10:48:14.168918 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [.], ack 233, win 511, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0
10:48:14.168945 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [F.], seq 78, ack 233, win 512, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0
10:48:14.168953 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [F.], seq 233, ack 79, win 512, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0
10:48:14.168967 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [.], ack 234, win 512, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0
```

9. virtual network/subnet assignment, public/private IP configuration, and associated Azure resources used to support the project simulation and packet-capture testing.

The screenshot displays the Microsoft Azure portal interface for a virtual machine named 'Zt'. The left sidebar shows the navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Monitor, Resource visualizer, Connect, Networking, Settings, Availability + scale, Security, Backup + disaster recovery, Operations, Monitoring, Automation, and Help. The main content area is divided into several sections:

- Essentials:** Shows basic information about the VM, including its resource group (Zt_group), status (Running), location (North Central US), subscription (Azure for Students), and subscription ID.
- Properties:** A detailed list of VM specifications such as Computer name (Zt), Operating system (Linux (ubuntu 24.04)), VM generation (V2), VM architecture (x64), Agent status (Ready), Agent version (2.15.1.3), Hibernation (Disabled), Host group, Host, Proximity placement group, Colocation status (N/A), Capacity reservation group, and Disk controller type (SCSI).
- Networking:** Displays network configuration, including the public IP address (23.101.174.187), private IP address (10.0.0.4), virtual network/subnet (Zt-vnet/default), and DNS name (Not configured).
- Size:** Shows the VM size (Standard B2ats v2) and its specifications (2 vCPUs, 1 GiB RAM, 2 threads per core).
- Source image details:** Lists the source image publisher (canonical), source image offer (ubuntu-24-04-lts), and source image plan (server).
- Disk:** Provides information about the OS disk (Zt_OsDisk_1_c9381425943d4906b8be4d9ddab1b842), encryption at host (Disabled), Azure disk encryption (Not enabled), ephemeral OS disk (N/A), and data disks (0).
- Auto-shutdown:** Indicates that auto-shutdown is not enabled.
- Health monitoring:** Shows that health monitoring is not enabled.
- Extensions + applications:** Lists any installed extensions or applications.

10. This screenshot shows the Python-based Zero Trust simulation running successfully inside the Azure Ubuntu VM. The simulation evaluated 13 access requests using identity, role, device

compliance, source segment, resource sensitivity, and monitoring rules. All 13 policy tests passed, with 3 requests allowed, 10 requests denied, and 13 alerts generated.

```
Last login: Sat May  9 22:06:17 2026 from [REDACTED]
azureuser@Zt:~$ cd ~/zt_lab
azureuser@Zt:~/zt_lab$ ls -lh src
total 4.0K
-rw----- 1 azureuser azureuser 125 May  9 22:28 zero_trust_simulation.py.save
azureuser@Zt:~/zt_lab$ rm src/zero_trust_simulation.py.save
azureuser@Zt:~/zt_lab$ nano src/zero_trust_simulation.py
azureuser@Zt:~/zt_lab$ python3 src/zero_trust_simulation.py
Zero Trust simulation complete
Total access requests: 13
Allowed requests: 3
Denied requests: 10
Policy tests passed: 13/13
Alerts generated: 13
Results saved in: /home/azureuser/zt_lab/results
azureuser@Zt:~/zt_lab$
```

11. This screenshot shows the output files generated by the Zero Trust simulation. The program created `access_log.csv`, `alerts.csv`, and `test_summary.csv`, which provide structured evidence of access decisions, monitoring alerts, and policy test outcomes.

```
azureuser@Zt:~/zt_lab$ ls -lh results
total 12K
-rw-rw-r-- 1 azureuser azureuser 3.4K May  9 22:38 access_log.csv
-rw-rw-r-- 1 azureuser azureuser 3.0K May  9 22:38 alerts.csv
-rw-rw-r-- 1 azureuser azureuser 836 May  9 22:38 test_summary.csv
```

12. This screenshot shows the `test_summary.csv` output. Each test case lists the access scenario, expected decision, actual decision, and policy result. The output confirms that the simulated Zero Trust policy engine correctly allowed authorized requests and denied unauthorized or risky requests.

```
azureuser@Zt:~/zt_lab$ cat results/test_summary.csv
request_id,scenario,expected_decision,actual_decision,policy_result
T01,HR user accesses HR records from compliant device,ALLOW,ALLOW,PASS
T02,HR user attempts Finance database access,DENY,DENY,PASS
T03,Finance user uses noncompliant device,DENY,DENY,PASS
T04,Admin accesses admin console from trusted segment,ALLOW,ALLOW,PASS
T05,Guest attempts internal HR access,DENY,DENY,PASS
T06,High-risk unmanaged device attempts admin access,DENY,DENY,PASS
T07,Valid HR user attempts access from wrong segment,DENY,DENY,PASS
T08A,Repeated failed attempt 1,DENY,DENY,PASS
T08B,Repeated failed attempt 2,DENY,DENY,PASS
T08C,Repeated failed attempt 3,DENY,DENY,PASS
T09,Device remains noncompliant after prior denial,DENY,DENY,PASS
T10,Guest accesses public portal,ALLOW,ALLOW,PASS
T11,Inactive account attempts access,DENY,DENY,PASS
```

13. This screenshot shows the `alerts.csv` output created by the monitoring engine. The alerts document policy denials and repeated denied access attempts, demonstrating how the simulation supports continuous monitoring and flags suspicious access behavior.

```
azureuser@Zt:~/zt_lab$ cat results/alerts.csv
timestamp,request_id,severity,alert_type,message
2026-05-09T22:38:40.734972+00:00,T02,MEDIUM,POLICY_DENIAL,Access denied for u_hr to FINANCE_DB: User role is not authorized for this resource; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.734981+00:00,T03,MEDIUM,POLICY_DENIAL,Access denied for u_fin to FINANCE_DB: Device is not compliant
2026-05-09T22:38:40.734993+00:00,T05,MEDIUM,POLICY_DENIAL,Access denied for u_guest to HR_RECORDS: User role is not authorized for this resource; MFA is required but not enabled; Managed device is required; Encrypted device is required for sensitive resource; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.734998+00:00,T06,MEDIUM,POLICY_DENIAL,Access denied for u_guest to ADMIN_CONSOLE: User role is not authorized for this resource; MFA is required but not enabled; Managed device is required; Device is not compliant; Encrypted device is required for sensitive resource; Device risk score is too high; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735003+00:00,T07,MEDIUM,POLICY_DENIAL,Access denied for u_hr to HR_RECORDS: Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735006+00:00,T08A,MEDIUM,POLICY_DENIAL,Access denied for u_guest to FINANCE_DB: User role is not authorized for this resource; MFA is required but not enabled; Managed device is required; Device is not compliant; Encrypted device is required for sensitive resource; Device risk score is too high; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735011+00:00,T08B,MEDIUM,POLICY_DENIAL,Access denied for u_guest to FINANCE_DB: User role is not authorized for this resource; MFA is required but not enabled; Managed device is required; Device is not compliant; Encrypted device is required for sensitive resource; Device risk score is too high; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735017+00:00,T08B,HIGH,REPEATED_DENIALS,Device d_unknown triggered repeated denied access attempts
2026-05-09T22:38:40.735017+00:00,T08C,MEDIUM,POLICY_DENIAL,Access denied for u_guest to FINANCE_DB: User role is not authorized for this resource; MFA is required but not enabled; Managed device is required; Device is not compliant; Encrypted device is required for sensitive resource; Device risk score is too high; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735017+00:00,T08C,HIGH,REPEATED_DENIALS,Device d_unknown triggered repeated denied access attempts
2026-05-09T22:38:40.735022+00:00,T09,MEDIUM,POLICY_DENIAL,Access denied for u_fin to FINANCE_DB: Device is not compliant
2026-05-09T22:38:40.735028+00:00,T11,MEDIUM,POLICY_DENIAL,Access denied for u_disabled to HR_RECORDS: Inactive user account; Device is not assigned to requesting user
2026-05-09T22:38:40.735028+00:00,T11,HIGH,REPEATED_DENIALS,Device d_hr_laptop triggered repeated denied access attempts
```

14. This screenshot shows the beginning of the `access_log.csv` file. The access log records each request timestamp, scenario, user, device, source segment, target resource, access

decision, expected decision, pass/fail result, and explanation. This provides traceable evidence for how the policy engine evaluated each access attempt.

```
azureuser@Zt:~/zt_lab$ head -5 results/access_log.csv
timestamp,request_id,scenario,user_id,device_id,source_segment,resource_id,decision,expected_decision,policy_result,reason
2026-05-09T22:38:40.734959+00:00,T01,HR user accesses HR records from compliant device,u_hr,d_hr_laptop,HR,HR_RECORDS,ALLOW,ALLOW,PASS,Access approved by Zero Trust policy
2026-05-09T22:38:40.734972+00:00,T02,HR user attempts Finance database access,u_hr,d_hr_laptop,HR,FINANCE_DB,DENY,DENY,PASS,User role is not authorized for this resource; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.734981+00:00,T03,Finance user uses noncompliant device,u_fin,d_fin_bad,FINANCE,FINANCE_DB,DENY,DENY,PASS,Device is not compliant
2026-05-09T22:38:40.734989+00:00,T04,Admin accesses admin console from trusted segment,u_admin,d_admin,IT_ADMIN,ADMIN_CONSOLE,ALLOW,ALLOW,PASS,Access approved by Zero Trust policy
```